A.2 Prerequisites

To install the OpenTelemetry demo in Kubernetes, you will need a Kubernetes cluster:

- If you want to execute locally, some popular choices are
  - Docker Desktop (https://docs.docker.com/desktop/kubernetes/)
  - kind (https://kind.sigs.k8s.io/)
  - K3s (https://k3s.io/)
  - minikube (https://minikube.sigs.k8s.io/docs/)
  - Microk8s (https://microk8s.io/)
- Cloud providers have managed Kubernetes offerings, depending on where you typically work. Check out the following:
  - Amazon Elastic Kubernetes Service (EKS; https://aws.amazon.com/eks/)
  - Azure Kubernetes Service (AKS; https://learn.microsoft.com/en-us/azure/aks/)
  - Google Kubernetes Engine (GKE; https://cloud.google.com/kubernetes-engine)

You will also need the following CLI tools:

- kubectl—Install via the instructions in the Kubernetes docs (https://kubernetes.io/docs/tasks/tools/).
- eksctl—Install as described in the docs (https://github.com/eksctl-io/eksctl#installation).
- *Helm*—Install based on the docs (https://v3-1-0.helm.sh/docs/intro/install/).
- jq—Install via the docs (https://jqlang.github.io/jq/download/).
- `curl` *and* `watch`—Should be available on your system; otherwise, consult your Linux distro or Windows docs.

I recommend having at least two medium-sized nodes available in your data plane. For example, I'm using EKS with a three-node cluster that is using t2.large (https://instances.vantage.sh/aws/ec2/t2.large) EC2 instances (with two vCPUs and 8 GB of RAM per node).

To provision the EKS cluster, I'm using eksctl (https://eksctl.io/) with the config file otel-demo-cluster.yaml, shown in the following listing.

Listing A.2 EKS cluster configuration

```
apiVersion: eksctl.io/v1alpha5
kind: ClusterConfig
metadata:
  name: otel-demo
  region: eu-west-1
  version: "1.24"
iam:
  withOIDC: true
```

Make sure to set this to your region (here, in Dublin, Ireland, eu-west-1).

This will provision the OIDC provider so that you can attach IAM policies if you want to change the OpenTelemetry collector config, for example, to send metrics to Amazon Managed Service for Prometheus or traces to AWS X-Ray.